

1. **Define wave function. Starting with the general solution of Schrodinger's equation for 1-D case, as $\psi(x) = A\sin kx + B\cos kx$ where the notations have their usual meanings, derive the relation between total energy and normalized wave function.**

Solution:

Wave function

It is a complex mathematical expression that defines the nature of particle in quantum mechanical domain.

$$\psi = Ae^{\frac{-j}{\hbar}[Et - px]}$$

Given,

$$\psi(x) = A\sin kx + B\cos kx$$

$$\text{We know, } k^2 = \left(\frac{2m(E-V)}{\hbar^2}\right)$$

Where, A and B are some constants which we can find from boundary conditions.

We know from boundary;

$$\psi(x) = 0 \text{ for } x = 0$$

Here,

$$0 = A \cdot 0 + B \cdot 1$$

$$B = 0$$

Again,

$$\psi(x) = 0 \text{ for } x = L$$

$$0 = A\sin kL$$

Since, $A \neq 0$ then $\sin kL = 0 = \sin n\pi$

When $n = 1, 2, 3, \dots, \infty$

$$kL = n\pi$$

$$k = \frac{n\pi}{L}$$

Now, from equation (ii)

$$\psi(x) = A \sin \frac{n\pi}{L} x$$

To find the value of A above equation can be normalized as;

$$\int_0^L |\psi|^2 dx = 1$$

$$\int_0^L A^2 \sin^2 \frac{n\pi}{L} x dx = 1$$

$$\int_0^L A^2 (1 - 2\cos \frac{n\pi}{L} x) dx = 1$$

$$\frac{A^2}{2} x|_0^L = 1$$

$$A = \left(\frac{2}{L}\right)^{\frac{1}{2}}$$

The normalized wave function of the electron in the box is;

$$\psi_n = \left(\frac{2}{L}\right)^{\frac{1}{2}} \sin \frac{n\pi}{L} x$$

This is called Eigen's function and value is used to measure the total energy of the particle inside the box.

$$k^2 = \left(\frac{2m(E-V)}{\hbar^2}\right) = \frac{8\pi^2 m(E-V)}{h^2}$$

$$\frac{n^2 \pi^2}{L^2} = \frac{8\pi^2 m(E-V)}{h^2} \quad (k = \frac{n\pi}{L})$$

$$L = \frac{nh}{\sqrt{8m(E-V)}}$$

Now,

$$\psi_n = \left(\frac{2\sqrt{8m(E-V)}}{nh}\right)^{\frac{1}{2}} \sin \frac{\pi\sqrt{8m(E-V)}}{h} x$$

2. **What is Heisenberg's uncertainty principle? Using this principle prove that minimum kinetic energy of a particle confined in an infinite potential well of width L is given by $E = \frac{h^2}{8mL^2}$, where h and m have their usual meanings.**

Solution:

Heisenberg's Uncertainty Principle

It is impossible to determine both position and momentum of an electron simultaneously with perfect accuracy. If an experiment is designate to measure one of them exactly, the other will become uncertain and vice-versa.

Let Δx be the uncertainty to introduce in measuring the position and Δp be the uncertainty to introduce in measuring momentum. If we try to measure both at once then according to uncertainty principle.

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Where, $\hbar = \frac{h}{2\pi}$, h Plank's constant.

An electron confined in an infinite potential well, the wave function of the particle extends from $x = 0$ to $x = L$. So, the uncertainty in position is equal to the width L of the potential well. The momentum of the electron is either $p_x = \hbar k$ in $+x$ direction and $p_{-x} = -\hbar k$ in $-x$ direction.

Therefore total uncertainty in momentum is given by $2\hbar k$. When the electron is in ground state, $n = 1$ and $kL = \pi$. So, the uncertainty in momentum can be written as,

$$\Delta p_x = \frac{2\pi\hbar}{L}$$

Now, the product of uncertainties in position and momentum of electron can be found.

$$\Delta x \Delta p_x = L \frac{2\pi\hbar}{L}$$

$$\Delta x \Delta p_x = 2\pi\hbar$$

$$\Delta x \Delta p_x = h$$

As, $\Delta p_x = 2p_x$

$$p_x = \frac{h}{2L}$$

Also, we have

$$E = \frac{p^2}{2m}$$

$$E = \frac{h^2}{8mL^2}$$

Above equation is the minimum required energy of an electron confined in an infinite potential well.

- 3. The normalized wave function for electron in infinite potential is $\psi_n = \left(\frac{2}{L}\right)^{\frac{1}{2}} \sin \frac{n\pi}{L} x$, where symbols have their usual meaning. Find the expected value of position and momentum for the electron.**

Solution:

The normalized wave function for electron in infinite potential well is $\psi_n = \left(\frac{2}{L}\right)^{\frac{1}{2}} \sin \frac{n\pi}{L} x$.

The expectation value of the position can be found by direct integration,

$$\begin{aligned} \langle x \rangle &= \int_0^L \psi_n^* x \psi_n dx \\ &= \frac{2}{L} \int_0^L x \sin^2 \left(\frac{n\pi x}{L} \right) dx \\ &= \frac{1}{L} \int_0^L x \left[1 - \cos \left(\frac{2n\pi x}{L} \right) \right] dx \\ &= \frac{1}{L} \int_0^L x dx - \frac{1}{L} \int_0^L x \cos \left(\frac{2n\pi x}{L} \right) dx \end{aligned}$$

$$= \left[\frac{x^2}{2L} \right]_0^L - \frac{1}{2n\pi} \left[x \sin \left(\frac{2n\pi x}{L} \right) \right]_0^L + \frac{1}{2n\pi} \int_0^L \sin \left(\frac{2n\pi x}{L} \right) dx$$

$$= \frac{L}{2}$$

The expectation value of the momentum is

$$\langle p \rangle = -i\hbar \int_0^L \psi_n^* \frac{d\psi_n}{dx} dx$$

$$= -i\hbar \frac{2}{L} \int_0^L \sin \frac{n\pi}{L} x \frac{n\pi}{a} \cos \left(\frac{n\pi x}{L} \right) dx$$

$$= -i\hbar \frac{1}{L} \frac{n\pi}{a} \int_0^L \sin \left(\frac{2n\pi x}{L} \right) dx$$

$$= 0$$

4. What is tunneling in quantum mechanics? Explain with the necessary mathematical expression, the nature of wave function in different regions in case of tunneling.

Solution:

Tunneling is a quantum mechanical phenomenon when a particle is able to penetrate through a potential energy barrier that is higher in energy than the kinetic energy of the particle.

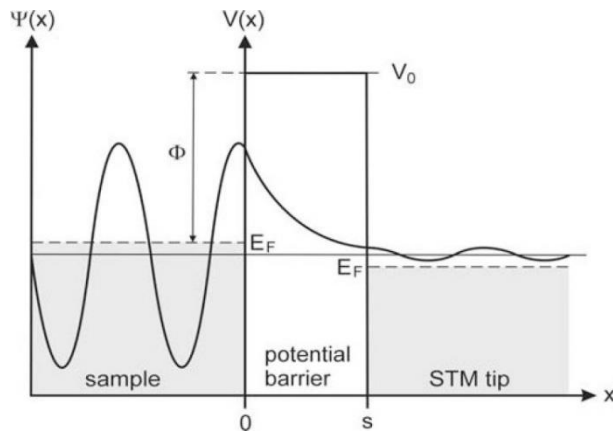


Fig : electron wave length in different region

In region 1, the Schrodinger's equation is

$$\frac{\partial^2 \psi_1}{\partial x^2} + \beta^2 \psi_1 = 0 \text{ ----- (i)}$$

And the solution is,

$$\psi_1 = Ae^{j\beta x} + Be^{-j\beta x} \text{ ----- (ii)}$$

In region 2, the wave function is,

$$\frac{\partial^2 \psi_2}{\partial x^2} + \alpha^2 \psi_2 = 0 \text{ ----- (iii)}$$

$$\psi_2 = De^{-j\alpha x} \text{ ----- (iv)}$$

Similarly, in region 3, the Schrodinger's equation is,

$$\frac{\partial^2 \psi_3}{\partial x^2} + \beta^2 \psi_3 = 0 \text{ ----- (v)}$$

And the solution is,

$$\psi_3 = Fe^{j\beta x} + Ge^{-j\beta x} \text{ ----- (vi)}$$

Where,

$$\beta^2 = \frac{2mE}{\hbar^2}$$

$$\alpha^2 = \frac{2m(E-V_0)}{\hbar^2}$$

$Ae^{j\beta x}$ = wave travelling in +ve x-direction

$Be^{-j\beta x}$ = wave reflected from potential barrier

$De^{j\alpha x}$ = exponential decrease wave that penetrates the barrier

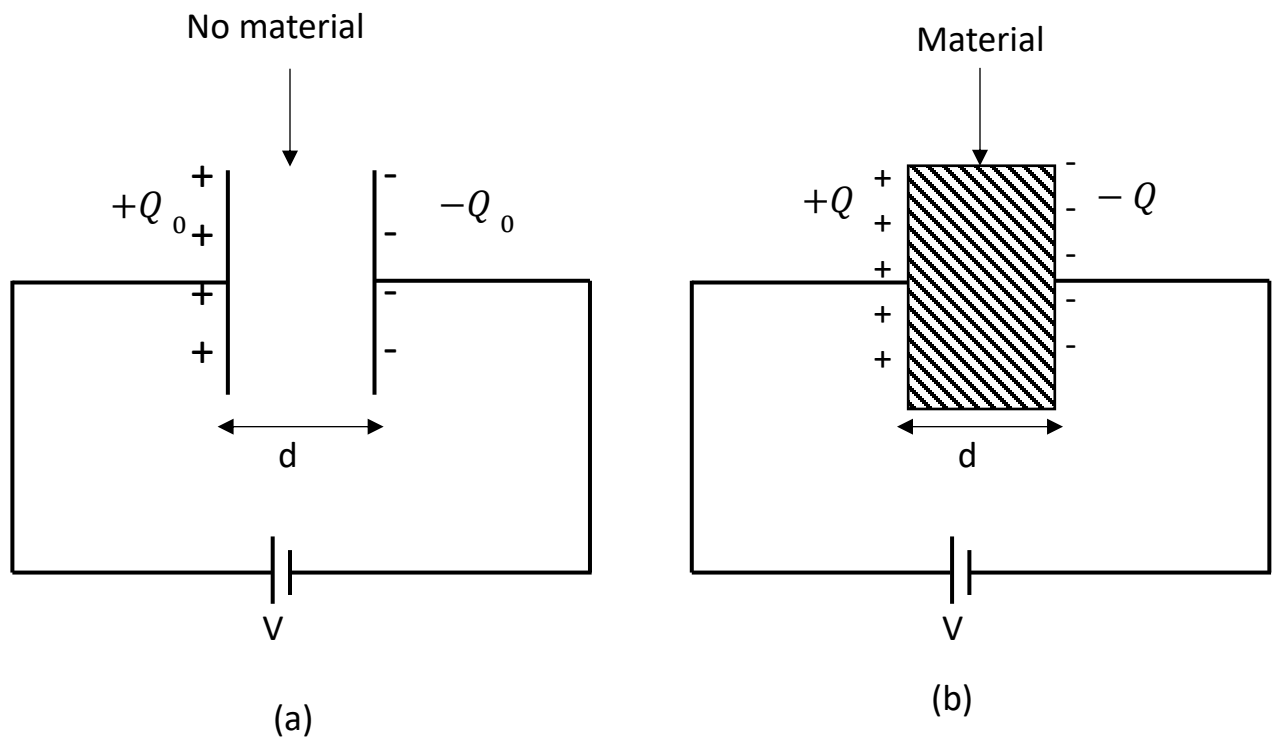
$Fe^{j\beta x}$ = wave travelling in +ve x-direction onwards

$Ge^{-j\beta x}$ = wave reflected from region 3 which practically and theoretically does not exist = 0

5. Define polarization. Show that “there will be no polarization if $\epsilon_r = 1$ ”.

Solution:

If electric field is applied to a medium made up of large number of atoms or molecules, the charges bound in each molecule will respond to applied field which will result in the redistribution of charges leading to a polarization of the medium. The electric polarization is defined as the dipole moment per unit volume.



Case (a) : Parallel plate capacitor without dielectric material

Case (b) : Parallel plate capacitor with dielectric material

For case (a) :

$$\text{Capacitance } (C_0) = \frac{Q_0}{V}$$

After insertion of dielectric between the plates, then the charges on the surface of plate get increased due to appearance of the charge on the surface of the dielectric near to the capacitor plates.

For case (b):

$$\text{Capacitance } (C) = \frac{Q}{V}$$

From case (a)

$$C_0 = \frac{Q_0}{V}$$

$$C_0 V = Q_0$$

$$V \times \frac{\epsilon_0 A}{d} = Q_0$$

$$\frac{\epsilon_0 V}{d} = \frac{Q_0}{A} = \sigma_0$$

From case (b)

$$C = \frac{Q}{V}$$

$$CV = Q$$

$$V \times \epsilon \frac{A}{d} = Q$$

$$\frac{V\epsilon}{d} = \frac{Q}{A} = \sigma$$

Where, σ = surface charge density

Now, from above equations

$$\frac{\sigma}{\sigma_0} = \frac{Q}{Q_0} = \frac{C}{C_0} = \epsilon_r$$

Therefore, dielectric constant or relative permittivity is defined as the increase in charge stored on capacitor plates due to insertion of dielectric medium compared to the charge without dielectric medium.

Increase in surface charge density when dielectric medium is inserted compared with that without dielectric is called polarization.

$$\text{Polarization}(P) = \sigma - \sigma_0 = \frac{Q}{A} - \frac{Q_0}{A}$$

$$P = \frac{\epsilon_0 \epsilon_r V}{d} - \frac{\epsilon_0 V}{d}$$

$$P = \frac{\epsilon_0 V}{d} (\epsilon_r - 1)$$

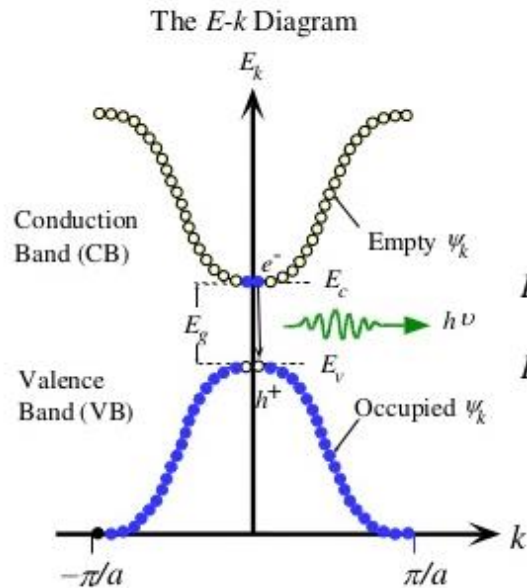
It is clear that there is no polarization if $\epsilon_r = 1$ from above equation.

6. Define effective mass of electron and explain about negative effective mass of electron.

Solution:

The electron in a periodic potential is accelerated relative to the lattice in an applied electric field as if its mass is equal to an effective mass. An electron in crystal may behave as if it had a mass different from free electron mass. There are crystals in which the effective mass of the carriers is much larger or much smaller than free electron mass. The effective mass may even be negative and zero.

The effective mass is given by, $m^* = \frac{\hbar^2}{d^2E/dk^2}$. So, mass depends upon curvature of energy in k-space (E-k diagram).



E-k diagram is parabolic in nature. For the lower portion of E-k diagram in conduction band curve is concave upward and that implies electron has positive mass, and increases with increase of k , attains a maximum value at the point of inflection. For further higher values of k , $\frac{d^2E}{dk^2}$ is negative as curve is concave downward which implies electron has negative effective mass. Physically, that is in the upper half of the band, the electron behaves as having negative mass or as behaving like particles with positive charges.

Suppose an electron starts at $k = 0$, when an electric field is applied, the wave vector increases linearly with time. Until the velocity reaches its maximum value, the electron is accelerated by the field, beyond this maximum velocity the same field produces a decrease in velocity, i.e the mass must become negative in the upper part of the band.

7. What type of magnetic material do you prefer for high frequency application and why?

Solution:

Soft magnetic materials are used for high frequency applications because of their following properties:

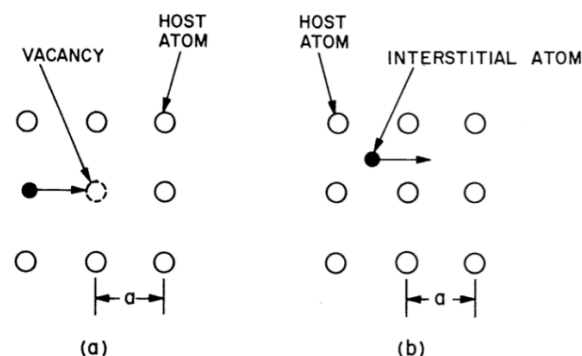
- a. High frequency implies magnetization and demagnetization in a faster rate. As a result, hysteresis losses are bound to take place. Soft magnetic materials have comparatively smaller area enclosed by hysteresis loop. Therefore, hysteresis losses are low.
- b. The magnetic permeability is also high, low coercive field. So, they can be easily magnetized in either direction which implies low eddy current losses.

8. What is diffusion? What are its types? Explain in detail about ion implantation process with neat sketch?

Solution:

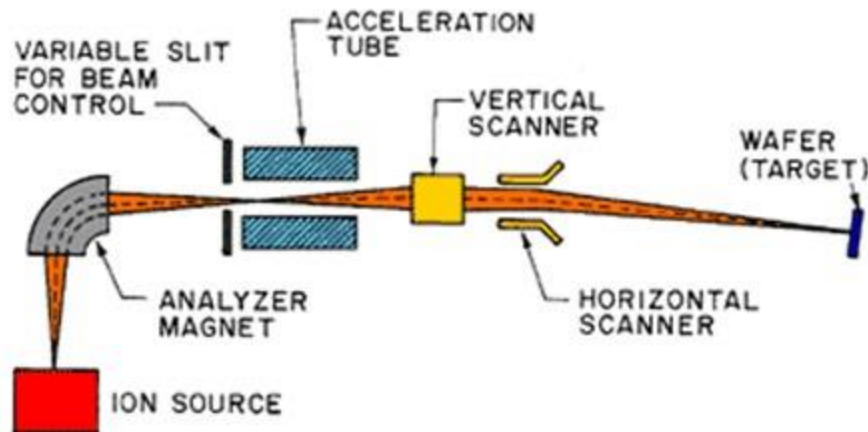
Diffusion in a semiconductor can be seen as a process of atomic movement of the dopant atoms in the crystal lattice. Basically there are two types of diffusion:

- a. Vacancy diffusion: At high temperature, the lattice atoms vibrate around the equilibrium lattice sites. There is a finite probability that a host atom can acquire sufficient energy to leave the lattice site and to become an interstitial atom thereby creating a vacancy. When a neighboring impurity migrates to the vacancy site, the mechanism is called vacancy diffusion.
- b. Interstitial diffusion: If an interstitial atom moves from one place to another without occupying a lattice site the mechanism is interstitial diffusion.



Ion Implantation

Ion implantation is a low temperature process by which ions of one element are accelerated into a solid target, thereby changing the physical, chemical, or electrical properties of the target.



Basic requirements for ion implantation systems are ion sources & means for their extraction, acceleration and purification. This is followed by beam detection and scanning prior to impingement of the substrate.

1. Ion source

Ion source includes compounds of species desired and means of their ionization. Gaseous materials are more preferred over solid materials since they do not need vaporization chamber. Effectiveness of ion source is measured as magnitude of ion current delivered to the accelerator and ultimately to target.

2. Accelerometer:

High energy is given to ion beams by passing them through a long column containing biased, annular, ring electrodes. These electrodes establish accelerating potential. Output end of column should be at ground potential for safety. The beam energy determines projected range of ion.

3. Mass analyzer:

Raw output from ion source from many species with contamination and hence purification of beam, to select desired implant species is important. It is carried out by means of mass analyzer that selects single species of interest. This can be done before or after the beam is accelerated. The mass separation technique allows handling variety of dopants in a single machine with freedom from contamination with each other. Most commonly employed technique uses a homogeneous field magnetic analyzer.

4. Beam scanner:

These are required for uniform coverage and ability to handle large number of slices in a single pump down.

Advantages:

1. Mass separation techniques can be used to obtain highly pure form of impurity atoms.
2. Wide range of dopant doses.
3. It can be carried out at room temperature.
4. Provides independent control of dose and penetration depth.

Disadvantages:

1. Equipment's are highly sophisticated and expensive.
2. Results in damage to conductor.
3. At high temp, annealing is needed.

9. What are advantages and disadvantages of ion implantation process over diffusion process?

Solution:

Advantages of ion implantation over diffusion process

- i. Low temperature process
- ii. Wide selection of masking materials
- iii. Less sensitive to surface cleaning procedures
- iv. Very fast (6" wafer can take as little as 6 seconds for a moderate dose)
- v. Complex profiles can be achieved by multi-energy implants

Disadvantages of ion implantation over diffusion process

- i. Very expensive equipment
- ii. At high dose value, throughput is less than diffusion.
- iii. Ions damage the semiconductor lattice. Damage can be corrected by annealing.
- iv. Very shallow and very deep doping are difficult or impossible.
- v. Masking materials can be knocked into the wafer creating unwanted impurities, or even destroying the quality of the interface.

10. Define diffusion length and obtain the continuity equation for holes and electrons.

Solution:

Diffusion length is the average length a carrier moves between generation and recombination. Semiconductor materials that are heavily doped have greater recombination rates and consequently, have shorter diffusion lengths. Higher diffusion lengths are indicative of materials with longer lifetimes and are, therefore, an important quality to consider with semiconductor materials.

The continuity equation which describes the distribution of electrons and holes when there is excess carrier generation, recombination and carrier movement.

Consider a differential length Δx of a semiconductor sample with area A in the yz -plane. The hole current density leaving the volume, $J_p(x + \Delta x)$, can be larger or smaller than the current density entering, $J_p(x)$, depending on the generation and recombination of carriers taking place within the volume. The net increase in hole concentration per unit time, dp/dt , is the difference between the hole flux per unit volume entering and leaving, minus the recombination rate. We

can convert hole current density to hole particle flux density by dividing J_p by electron charge e . The current densities are already expressed per unit area; thus dividing $J_p(x)/e$ by Δx gives the number of carriers per unit volume entering $\Delta x A$ per unit time, and $J_p(x + \Delta x)/e\Delta x$ is the number leaving per unit volume and time.

$$\frac{\partial \Delta p}{\partial t} \Big|_{x \rightarrow \Delta x} = \frac{1}{e} \frac{J_p(x) - J_p(x + \Delta x)}{\Delta x} - \frac{\Delta p}{\tau_p}$$

The term in left hand side gives the rate of hole build-up whereas the first term in right hand side represents increase in hole concentration in volume $A\Delta x$ per unit time, and the last term is recombination rate.

As, limit $\Delta x \rightarrow 0$

$$\frac{\partial p(x, t)}{\partial t} = -\frac{1}{e} \frac{\partial J_p}{\partial x} - \frac{\Delta p}{\tau_p}$$

The equilibrium concentration is independent of time. $P_0 = \text{constant}$. So,

$$\frac{\partial p}{\partial x} = \frac{\partial \Delta p}{\partial x}$$

Now, from the above two equations, we have

$$\frac{\partial \Delta p}{\partial x} = -\frac{1}{e} \frac{\partial J_p}{\partial x} - \frac{\Delta p}{\tau_p}$$

Above equation is called continuity equation for holes.

Similarly, for electrons we have

$$\frac{\partial n}{\partial x} = \frac{1}{e} \frac{\partial J_e}{\partial x} - \frac{\Delta n}{\tau_e}$$

Above equation is called continuity equation for electrons.

11. State Fick's law of diffusion.

Solution:

Fick's law of diffusion explains how molecules move from higher concentration to lower concentration region. A diffusion process that obeys Fick's laws is called normal diffusion or Fickian diffusion and those who don't are called Anomalous diffusion.

There are two laws that are interrelated i.e Fick's first law is used to derive Fick's second law which is similar to the diffusion equation.

Fick's First Law

"The molar flux due to diffusion is proportional to the concentration gradient".

$$J = D \frac{d\phi}{dx}$$

Where, J, D, ϕ and x are diffusion flux, diffusivity, concentration and position respectively.

Fick's Second Law

“The rate of change of concentration of the solution at a point in space is proportional to the second derivative of concentration with space”.

$$\frac{\partial \phi}{\partial t} = D \frac{\partial^2 \phi}{\partial x^2}$$

12. Why silicon is preferred for semiconducting materials than Germanium? Explain the Floating Zone process of crystal growth with necessary figure.

Solution:

The main reasons for silicon to be preferred over germanium are:

- The availability of silicon is higher than germanium
- The silicon diodes are more stable with change in temperature than germanium ones.
- The silicon diode has low reverse current compared to germanium.
- The silicon diode has a large reverse breakdown voltage of about 70-100V compared to germanium which has the reverse breakdown voltage around 50V
- The structure of germanium crystals will be destroyed at a higher temperature.
However, silicon crystals are not easily damaged by excess heat.

Floating Zone process

Float-zone silicon is a high-purity alternative to crystals grown by the Czochralski process. The basic idea in float zone (FZ) crystal growth is to move a liquid zone through the material. If properly seeded, a single crystal may result. The production takes place under vacuum or in an inert gaseous atmosphere. The process starts with a high-purity polycrystalline rod is held in a vertical position and is rotated.

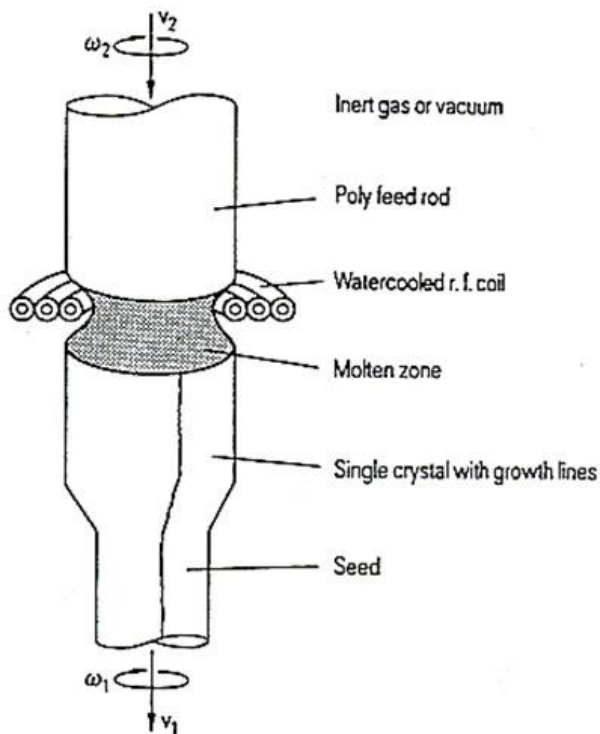
With a radio frequency the rod is partially melted. The seed is brought up from below to make contact with the drop of melt formed at the tip of the poly rod. As the molten zone is moved along the polysilicon rod, the molten silicon solidifies into a single Crystal and, simultaneously, the material is purified. FZ crystals are doped by adding the doping gas phosphine (PH₃) or diborane (B₂H₆) to the inert gas for n- and p-type, respectively.

Unlike CZ growth, the silicon molten Zone is not in contact with any substances except ambient gas, which may only contain doping gas. Therefore FZ silicon can easily achieve much higher purity and higher resistivity.

After the heating coils move over the whole polysilicon rod, it converts to a single crystal silicon ingot.

Advantages:

1. When extremely high purity silicon is required the growth technique of choice is float-zone method.
2. There is no need for a crucible and so there is lower melt contamination, especially oxygen and carbon which cannot be avoided in CZ crystal growth.



13. Short notes on

a. Seebeck effect

Solution:

A german physicist discovered that when two dissimilar metal wires are joined at two ends to form a loop, a voltage is developed in the circuit if the two junctions are kept at different temperatures. The pair of metals forming the circuit is called a thermocouple. The effect is due to conversion of thermal energy to electrical energy.

When heat is applied to one of the two conductors or semiconductors, heated electrons flow toward the cooler one. If the pair is connected through an electrical circuit, direct current flows through that circuit. The voltage produced by Seebeck effect are small, usually only a few microvolts per kelvin of temperature difference at the junction. If the temperature difference is large enough, some Seebeck-effect devices can produce a few millivolts.

Large arrays of Seebeck-effect devices can provide useful, small-scale electrical power if a large temperature difference is maintained across the junction.